

Homework 1: Demand and Exchange Economy

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Due: Sep 3, 2025, noon.

Demand and Income Expansion Path

For the utility function

$$u(x, y) = \ln(x + a) + \ln(y - b) \text{ s.t. } p_x x + p_y y = m.$$

for positive a and b , solve for the demand (note: the utility function is not well-defined *everywhere* on R_+^2)

Pick p_x and p_y . In the space of (x, y) plot points of x and y that this consumer will choose as you increase m . How does it change with a and b ?

Laspeyres and Paasche

To compare quantities consumed between two years, for instance to see if a representative family consumed more or less than last year, Laspeyres index suggests using prices of the last year, and Paasche index suggests using prices of the new year. Compare these approaches to EV and CV measures.

Indifference curves

For each case, plot indifference curves that are:

- monotone but not strictly monotone;
- convex but not strictly convex;
- convex but not monotone;
- monotone but not convex.

Exchange Economy Equilibrium

Consumer i has the utility function

$$u_i(x, y) = \alpha_i \ln(x) + (1 - \alpha_i) \ln(y).$$

There are n consumers, α_i are in $(0, 1)$. Consumer i 's endowment is $(X_i, Y_i) > 0$. Solve for the market equilibrium. Is it always the case that an increase in one of the X_i lowers good x 's price?